1. Data Collection Module
Ganache v7.4.0
Blockchain Testnet
(103,734 transactions)
Block generation
12s intervals
Mempool Monitor
(15 tx/s simulation)
Transaction
broadcast
Transaction Stream
(5 attack categories)
Raw transactions
2. Feature Engineering Module
Feature Extractor
(12 features)
2. Feature Engineering Module
Nonce Features
(conflict, gap)
Gas Features
(price, differential, volatility)
Timing Features
(frequency, interval)
Transaction Features
(value, data_size)
Z-Score Standardization
(μ=0, σ=1)
Bold edges = Primary data flow
Dashed edges = Internal module connections
Legend
80/20 split
82,987 train
3. Model Training Framework (Parallel Evaluation)
Classical ML
XGBoost
(269 trees, Bayesian opt)
96.48% acc
Random Forest
(200 trees)
96.48% acc
SVM (Linear+RBF)
(GridSearchCV)
61-82% acc
Deep Learning
MLP (12K params)
96.49% acc
1D-CNN (9K params)
96.46% acc
BiLSTM (79K params)
96.49% acc
Transformer (80K params)
96.49% acc
4. Model Selection & Statistical Validation
5-Fold Cross-Validation
(F1-weighted scoring)
Statistical
significance
McNemar's Test
(p<0.001 vs SVM)
Feature Ablation
(12 experiments)
XGBoost Selected
(96.48% acc, 0.0035ms)
Production
deployment
5. Production Deployment System
Real-Time Detector
(1,280 tx/s throughput)
5. Production Deployment System
SHAP Analysis
(global importance)
LIME Explanations
(instance-level)
Explainable
classifications
Alert System
(5 attack categories)